

SUMMARY

Enterprise-scale software engineer proficient in both application- and infrastructure-level development with expertise in SWE practices including Agile, DevOps, IaaS, CI/CD. Experienced in R&D of high-performance, low-latency systems. Strong foundation in mathematics, algorithms, and formal verification, passionate about architecting **maintainable**, **reliable**, and **large-scale** software systems.

WORK EXPERIENCE

Software Engineer → Senior SWE, Radin Bourse (Tehran, Iran) 2018 - 2023

- Designed and developed an ultra-low-latency trading engine for tehran stock exchange using LMAX Disruptor. Achieving $\sim 50k$ TPS with $20 \mu s$ latency for sequential order processing.
- Led design and migration of an on-permise/bare-metal stack to an IaaS solution using Ansible, MaaS, Docker, Terraform, Nomad, Vault, and Consul.

Backend Developer, Tapsi (Tehran, Iran) 2017 - 2018

Developed and deployed backend features for Tapsi, the second largest online taxi platform in Iran.

Software Developer, chibaki.ir (Tehran, Iran) 2016 - 2017

Independently designed, developed, and deployed the backend of an online platform for connecting clients to professionals, using Node.js.

EDUCATION

2024 - Present Ph.D. in Computer Science, Cheriton School of Computer Science, University of Waterloo
2019 - 2022 M.Sc. in Software Engineering, School of ECE, University of Tehran
2014 - 2019 B.Sc. in Computer Engineering, School of ECE, University of Tehran

PUBLICATION

2024 **Seyhani A**, Zhao J, Gupta A, Walker D, Arashloo MT. [Buffy: A Formal Language-Based Framework for Network Performance Analysis](#). 23rd ACM Workshop on Hot Topics in Networks.

NOTABLE PROJECTS

Count Buffy (2025) Verification framework built on Z3 for checking performance properties of network schedulers, supporting realistic buffer sizes while having efficient verification time.
[Estrest](#) (2022) A tool for checking actual causality using event-structure semantics of programs.
[APHTTP](#) (2017) A light-weight Express.js inspired HTTP web-server and template rendering implemented with pure C++, to enable simple web development with C++.

SKILLS

Languages	Java, C++, Python, C, SQL, Javascript
Frameworks & Libraries	Spring Frameworks (Boot, Data, Security), JMH, Lombok, Cucumber, Hibernate, Z3, Node.js NumPy, Pandas, Scikit-learn
Tools & Technologies	Ansible, Hazelcast, Maven, Terraform, Packer, Vault, Nomad, Consul, Docker, MySQL, MongoDB, JFR, GitLab CI, NGINX, mCRL2, , Linux, Git, AWS, GCP, Prometheus, Grafana
Additional Experience	Kafka, OpenCV, RabbitMQ, ELK Stack